



AgSpray Atomization Characterization using Deep Learning

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Abstract

Predicting droplet size distribution as a function of chemical composition, input pressure, nozzle geometry, and atmospheric conditions using algebraic approach is extremely complicated due to the number of variables involved, especially for agricultural spray applications. In our previous work, a low-cost shadowgraph technique was used to capture the spray pattern from six different standard ASABE nozzles (ranging from extremely coarse 6515 type to very fine 11001 nozzle type) at different inlet pressures, along with various combinations of adjuvant mixtures. The resultant images were then fed into convolutional neural network (CNN) architecture to classify the different nozzle types and chemical compositions. In this work, we extend the algorithm to predict different pressure distributions of the spray under each nozzle type and chemical compositions for two different regions of interest (ROI). The nozzle pressures were able to be predicted from the shadowgraph images with a mean squared error of 4.2 psi. An iPhone camera was also used to capture images of the spray from different nozzles which was used to train a similar deep learning algorithm for classification with an overall prediction accuracy of 100%.

I. Nomenclature

<i>ASABE</i>	<i>American Society of Agricultural and Biological Engineers</i>
<i>CNN</i>	<i>Convolutional neural network</i>
<i>EPA</i>	<i>Environmental Protection Agency</i>
<i>LSWT</i>	<i>Low Speed Wind Tunnel</i>

II. Introduction and Background

Spray drift is one of the major concerns in agricultural industry which affects nozzle manufacturers, chemical manufacturers and more importantly, farmers who are the applicators of the chemicals. Over the course of many years, spray drift issues have led to loss of lives and several lawsuits between the three dependent branches – farmers, nozzle manufacturers and chemical manufacturers [1]. Three-fourths of agriculture-related complaints investigated by the Ohio Department of Agriculture involved drift issues. Two-thirds of the total complaints in a five-year period brought to the attention of Iowa Department of Agriculture were related to drift problems and one-third of court cases due to spray misapplications reported by a major insurance company involved drift damages [2]. With the advent of genetically modified crops, the use of non-selective herbicides will be restricted because even a small amount of these herbicides can cause serious damage to off-target plantations. For example, applications of 2,4-D and Dicamba at levels as low as 1/300th of the field rate for soybeans caused statistically significant field loss in tomato crops. Drift of either herbicide caused a 17-77% reduction in the marketable fruit when applied a 1/100th of the field rate [3]. The number of herbicide applications is increasing substantially each year, with the U.S. alone increasing herbicide use by 500% between 2000 and 2014 [4]. In 2017, the use of Dicamba on Monsanto's Dicamba-resistant crops made headlines when the uncontrolled drift affected millions of acres of farms from Arkansas to North Carolina [5]. With the increasing number of spray applications each year, the likelihood of off-target movement of the sprays is of great concern.

Despite the increased use and risk, it is believed that one of the main challenges has to do with how the chemicals are applied with a nozzle type by farmers. For instance, it is customary for farmers to mix chemicals

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(pesticides, adjuvants, etc.) during a spray application that has not been tested in a wind tunnel for spray drift potential in order to save time, money, and effort. As such, the atomization of the fluid changes for the rated nozzle type and pressure in a manner that inadvertently increases the drift potential under certain conditions resulting in spray drift and consequential harm. Therefore, the goal of this research is to develop technology that can accurately classify any given agricultural spray into 6 categories: extremely coarse, very coarse, coarse, medium, fine, and very fine. The droplet size distributions for these different nozzles are shown in Table 1.

Table 1 Droplet diameter distributions from different ASABE nozzles

Nozzle	Rated Pressure (psi)	Dv(10) (μm)	Dv(50) (μm)	Dv(90) (μm)
11001 (Very Fine)	65.3	64.7	141.8	245.7
11003 (Fine)	43.5	110.9	247.6	412.0
11006 (Medium)	29.0	161.2	352.0	561.2
8008 (Coarse)	36.3	201.3	442.3	739.3
6510 (Very Coarse)	29.0	243.1	528.6	854.2
6515 (Extremely Coarse)	21.7	337.1	693.6	1088.5

Dv(10) indicates the 10th percentile by volume of the spray, Dv(50) indicates the 50th percentile by volume of the spray, and Dv(90) indicates the 90th percentile by volume of the spray. Droplets less than 100 μm are especially prone to spray drift. The measurements in Table 1 are determined by the use of Spraytech Laser Diffraction instrument in the University of Dayton Low Speed Wind Tunnel. However, these types of measurements are extremely difficult to obtain in real-time during the actual spray application by the farmers.

Since the atomization of the droplets changes when different chemicals are mixed, standard spray drift prediction models such as AgDisp and AgDrift does not work. Moreover, chemical and nozzle companies introduce several new products every year, especially shear thickening adjuvants/chemicals in an effort to minimize spray drift which makes the existing prediction models obsolete [6]. Hence, a new method to characterize and predict the spray drift is needed for chemical mixtures which can adapt to new nozzle design and new viscoelastic chemical compositions. Therefore, the current work takes advantage of recent advances in vision-based machine learning and deep learning technologies where algorithms are currently being used across many applications and disciplines which include but not limited to medical image analysis, face recognition, object detection, etc. One primary advantage of machine learning algorithms is continuously assessing and learning from spray atomization images and recognizing features/patterns (such as fluid ligament formation, break up, droplet sizes, etc.) that are essential for identification of the chemical composition and nozzle type. In addition, the class activation mapping results [7] for our deep learning algorithms were investigated to provide further insights about the prediction to the user. The speed at which the machine learning consumes and predicts data allows it to discover complex relationships and produce real-time data and predictions that are unparalleled by any other statistical, algebraic, or computational models.

The myriad problems associated with understanding, modeling, and predicting spray drift is due to the number of variables or factors that significantly influence it. Fluid properties such as temperature, viscosity, surface tension, volatility and viscoelasticity play a significant role in the atomization process. These properties when coupled with nozzle properties such as nozzle type, fan angle, inlet pressure, orifice size, etc. make the atomization process even harder to understand, let alone predict. Even though there are several dependent variables, controlling the droplet sizes that comes out of a nozzle has significant impact on controlling the spray drift. The different agricultural chemical compositions adversely change the properties of the fluid (analogous to adding a drop of soap in water), especially its viscoelasticity. The atomization process which controls the droplet size distribution depends on the fluid properties and the nozzle properties which are the main areas of focus in the proposed research.

An example shadowgraph image of a spray from an agricultural nozzle is shown in Figure 1 [8]. It can be observed that when the fluid leaves the nozzle, it forms ligaments and then these ligaments breakup into smaller droplets. This process is known as atomization, which is highly dependent on the properties of the fluid and the nozzle. Most characterization of atomization process has only been conducted on Newtonian fluids, such as water and oil – relatively thin, homogenous liquids that don't include any long polymers. It is customary for agricultural applications

to use chemical agents (fluid thickening agents/adjuvants) to reduce spray drift which changes the viscoelastic properties of the fluid, which in turn highly influences the atomization process.

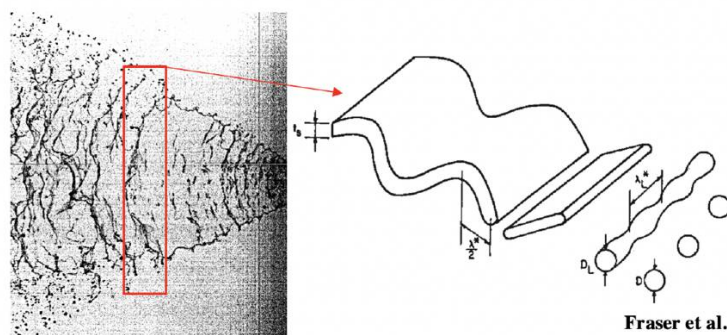


Figure 1 Shadowgraph image of the spray and the breakup mechanism proposed by Fraser et al. [8]

Newtonian fluids produced a narrower range of droplet sizes, regardless of the type of experiment performed, whereas the viscoelastic fluids had broader distributions, generating larger numbers of both big and small drops. When they are sprayed, the viscoelastic fluids are shown to create long ligaments, or string-like projections (as seen in Figure 1), then eventually break apart into droplets. Such information is vital in understanding the atomization process which can be gained through shadowgraph technique. The employment of this technique is conventionally expensive, however, a low cost low-speed shadowgraph technique was utilized to identify the features in the atomization process as a function of various fluid and nozzle properties. The atomization process leads to a droplet size distribution which can influence the rate of on-target deposition as well as canopy penetration, with smaller droplets providing better deposition and canopy penetration than larger ones [9]. However, droplet sizes below 100 μm in diameter are the most prone to spray drift [10], [11]. Along with the droplet size, droplet velocity also affects the structure of the spray deposits and the “driftability” of the droplets [12].

Using the shadowgraph technique, the relationship between ligament break up, droplet sizes, droplet velocity and structure can be understood for a viscoelastic chemical which will also benefit other spray applications. There are many theoretical and semi-theoretical models for predicting the atomization of liquid jets issuing from circular orifices into air as summarized by Lefebvre [13]. However, for flat fan nozzles, which are used in agricultural applications to spray pesticides from ground boom, the complex geometry of the nozzle makes it difficult to find simplifying assumptions. As a result, most droplet size analyses rely on empirical correlations and databases. Nonetheless, there have been some attempts at theoretical atomization models, although none have thus far found common use. To date, none of the works has resulted in a suitable model to predict the droplet size for flat-fan nozzles. Therefore, it is imperative that a robust prediction model be developed which 1) Provides reasonable accurate predictions of atomization process occurring from a given chemical composition and a nozzle type. 2) Is *adaptable* to the new nozzle designs and chemical compositions which are manufactured by nozzle and chemical companies.

III. Prior Work

In our prior work [14], advanced machine learning algorithms were trained by high speed shadowgraphy images of the spray pattern from different nozzles as well as images obtained by a traditional cell phone camera to classify a given spray pattern into the above six categories which will help farmers predict the spray drift potential of any given nozzle-chemical-combinations. The machine learning architecture used in our prior work is shown in Figure 2 to identify the chemical composition and nozzle type from the shadowgraph images. An input shadowgraph image of known chemical compositions sprayed from a known nozzle will be fed into the algorithm to first identify the mixture ratio of chemical composition and then to nozzle identification. The classification of chemical composition was done through the ResNet transfer learning approach as shown in Figure 2. The same transfer learning approach was used for nozzle identification as well.

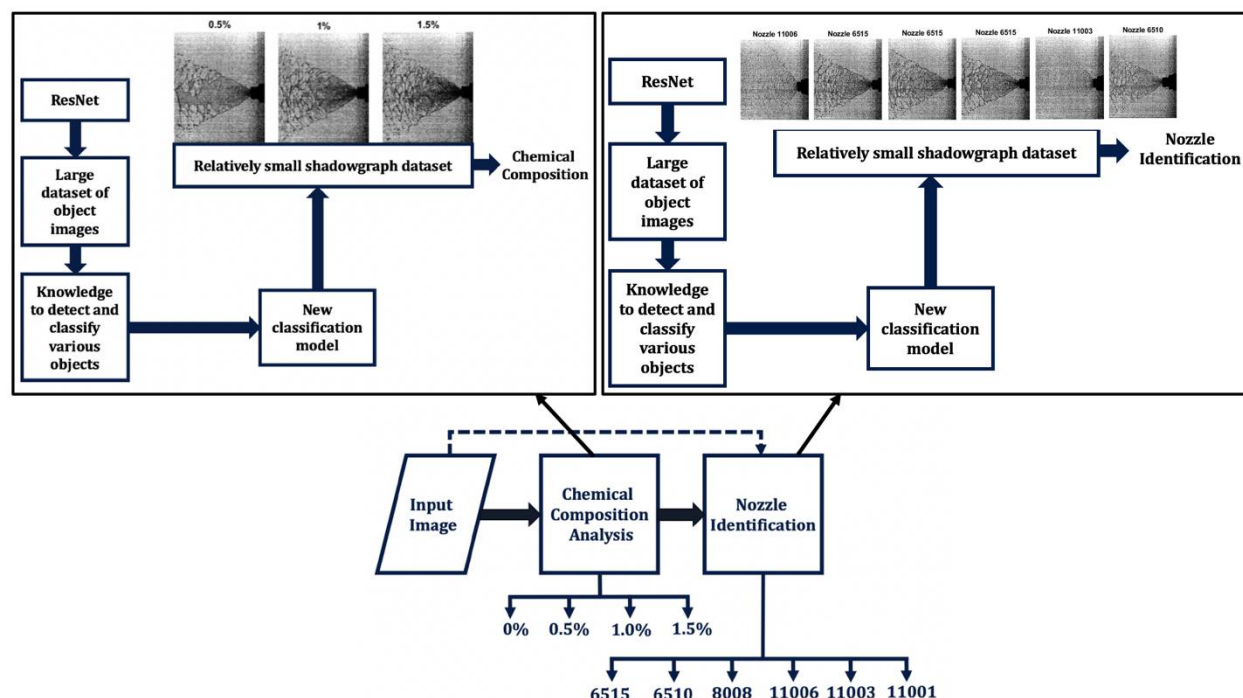


Figure 2 Machine learning architecture used to predict chemical composition and nozzle type [14]

Using the machine learning architecture shown in Figure 2, the chemical composition algorithm based on transfer learning approach gave an overall prediction accuracy of 98.4%. When using similar ResNet transfer learning approach, the nozzle identification for an identified chemical composition was able to be predicted with an accuracy of 99.9%. Apart from the predictions, the Class Activation Mapping (CAM) results were investigated to understand the underlying reasoning behind the algorithm's classification. An example CAM results from our prior work for a chemical composition of 1.5% is shown in Figure 3.

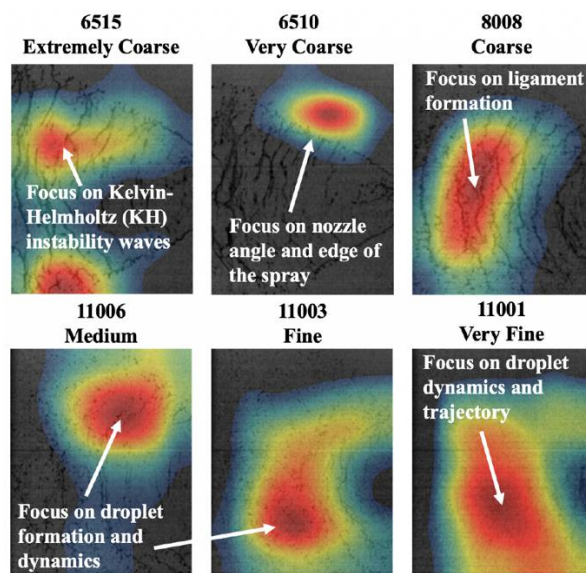


Figure 3 Mean CAM results for nozzle identification for 0% chemical composition (Water)

In the current work, the same architecture was extended to predict the pressure with which the spray was ejected out of the nozzle for an identified chemical and nozzle type. Pressure plays a crucial role in determining the droplet

distributions from the spray regardless of the type of nozzle used and the chemical composition. Generally, increasing the nozzle input pressure results in more finer droplets and hence the drift potential. Therefore, it is crucial to be able to predict the pressure at which the chemicals are sprayed from the shadowgraph image. Given that the primary purpose of this research is to help farmers classify the spray of a given tank mix into 6 main categories, one of the major disadvantages of the current method is that it is extremely difficult to capture high speed shadowgraph images of the spray from a tractor boom in-field. However, almost every farmer has access to a phone camera which can be used to capture images or videos of the spray from the nozzle. Therefore, a standard iPhone camera was used to capture videos of water being sprayed from different nozzle types and a similar deep learning architecture as shown in Figure 2 was developed to predict the nozzle type from spray pattern.

IV. Methodology

A. Data Acquisition

All experiments were conducted in the spray test chamber at the University of Dayton Low Speed Wind Tunnel laboratory. The spray test chamber houses the nozzle on a vertical traverse, a constant pressure pump, and several stage filters to filter any chemicals that are sprayed in the chamber. The two windows on either side of the test chamber allows to place the laser diffraction instrument as well as setup the necessary equipment for shadowgraphy technique. The maximum wind speed range with the spray drift chamber in the wind tunnel is 5 to 20 mph.



Figure 4 Spray Test Chamber at the University of Dayton Low Speed Wind Tunnel (UD-LSWT)

A 40W 1100 LM LED panel was used as a light source for the shadowgraphy technique along with a PCO 1200 camera for various chemical compositions, nozzle types and pressures. The chemical used in the current investigation was the InterLock adjuvant mixed with water at different percentages. The test matrix of the experiment was shown in Table 2 where all possible combinations of each variable was tested.

Table 2 Test matrix for pressure identification and nozzle identification by iPhone10 camera

Nozzle Type	InterLock Adjuvant Composition	Nozzle inlet pressure variation from the rated pressure (%)	Freestream velocity (m/s)	Image Acquisition Cameras
11001, 11003, 11006, 8008, 6510, 6515	0%, 0.5%, 1.0%, 1.5%	90, 100, 110	6	PCO 1200 HS
11001, 11003, 11006, 8008, 6510, 6515	0%	100	6	iPhone10

For the high-speed shadowgraph images, the field of view was calibrated to ensure the spray plane was properly focused on. During data collection, the camera was kept in the same location with the focus unchanged because every

nozzle was spraying in the same plane. To ensure that the camera produced visible images with minimal blur from the droplets, the exposure was selected to be 20 to 25 microseconds. In combination with using the full available frame of view, most trials were taken at a frame rate of about 500 frames per second. A 55 mm lens was used on the camera, with a fully open aperture to allow as most light as possible onto the sensor. The LED panel light was set to 100% power with a color temperature of 5600K for all trials. Trails were collected by starting the pump and allowing it to run until the operating pressure stabilized; the pressure was varied to match the required percentage stated in the testing matrix. Approximately 500 images were collected for each case.

For the spray image acquisition using an iPhone10 camera, the spray was illuminated with a 200mW Nd:YAG laser. The experimental setup for iPhone camera image acquisition of the spray is shown in Figure 5.

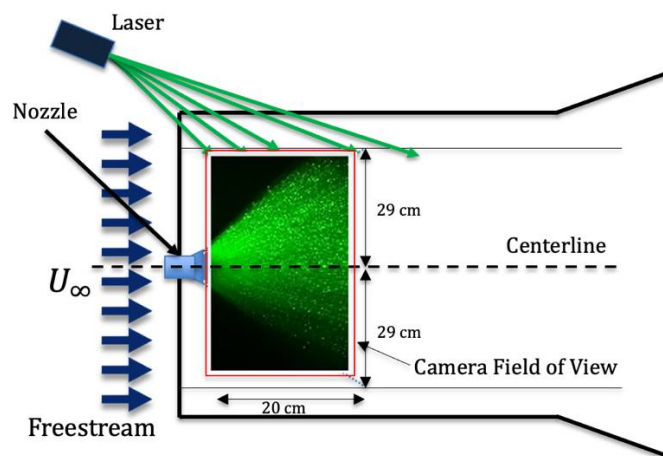


Figure 5 Experimental setup of spray image acquisition using iPhone 10 camera

B. Image Pre-Processing for Shadowgraph Images

The images obtained from shadowgraphy was first converted to gray scale. Then, three different regions of interest (ROI) were identified as an input to the machine learning classifier for each case. The three different ROIs are illustrated in Figure 6.

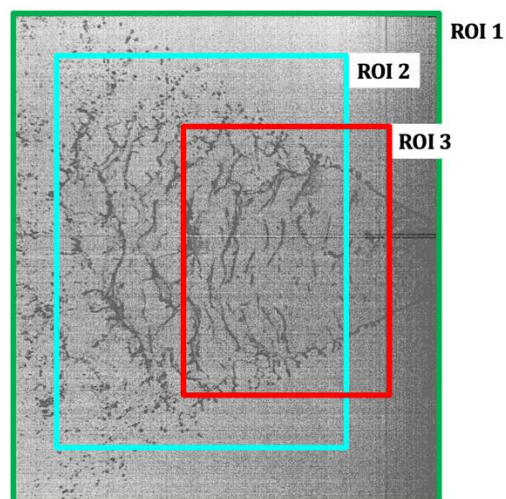


Figure 6 Three different regions of interest (ROIs) for the shadowgraphy images

C. CNN Approach for Nozzle Type Prediction using iPhone

In this section, we present a compact CNN architecture for the classification of images captured using iPhone into different nozzles. At first, we spatially re-sample all the images to 256×256 . Figure 7 presents the compact CNN architecture presented for the classification of iPhone images. Each convolutional layers present the architecture is comprised of 3×3 convolution operation, batch normalization, Rectified Linear Unit (ReLU) and maximum pooling layer of window size 2×2 with a stride of 2 as shown in Figure 8. Number of convolution filters present in each convolutional layer is present in Table 3. Hyper parameters are solely decided based on the performance on the validation dataset. We choose a mini batch size of 64 with Adam optimization algorithm and an initial learning rate of 0.0001.

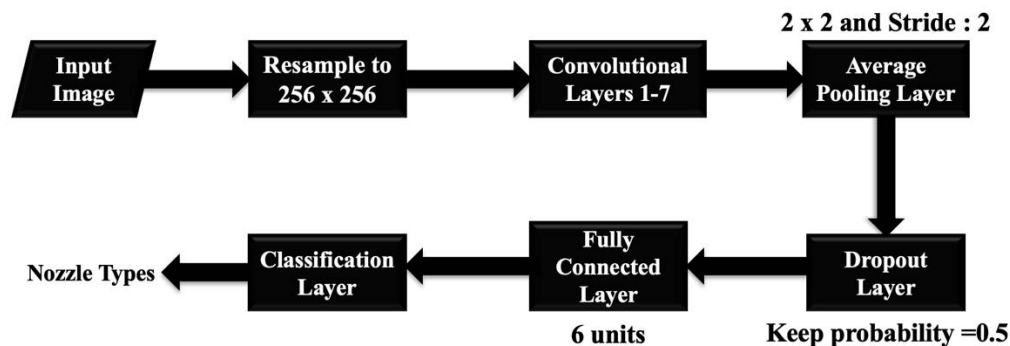


Figure 7 Block level diagram of the CNN approach used for nozzle identification from iPhone camera

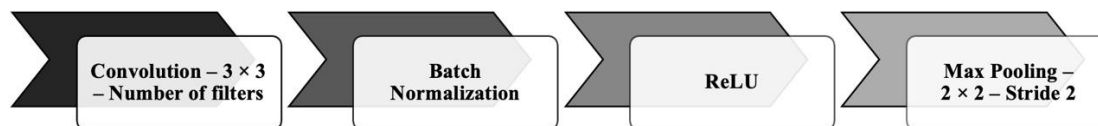


Figure 8 Convolutional Layer Structure

Table 3 Number of convolution filters present in each layer

Convolutional Layer #	Number of Filters
#1	16
#2	32
#3	64
#4	128
#5	256
#6	256
#7	512

D. Pressure Identification from Shadowgraph Images

In order to estimate the pressure from shadowgraph images, we tweak the CNN architecture presented for nozzle type prediction. We estimate the pressure % using a regression approach rather than the classification approach presented for nozzle identification. We alter the fully connected units from 6 to 1 and change the classification layer to regression layer as we are trying to estimate the exact pressure value from a given image. Figure 9 presents the modified CNN approach for estimating the pressure level. Number of convolution filters is the same as presented in Table 3. Figure 10 presents the distribution of the dataset utilized for this study.

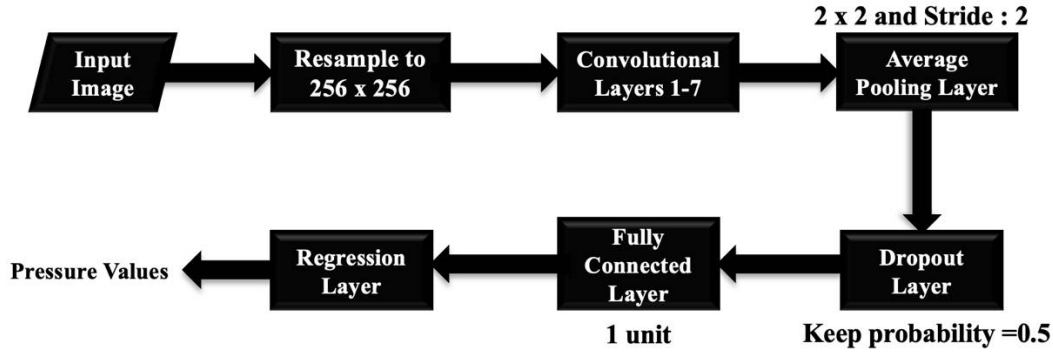


Figure 9 Block level diagram of the CNN approach used for pressure estimation on shadowgraph Images

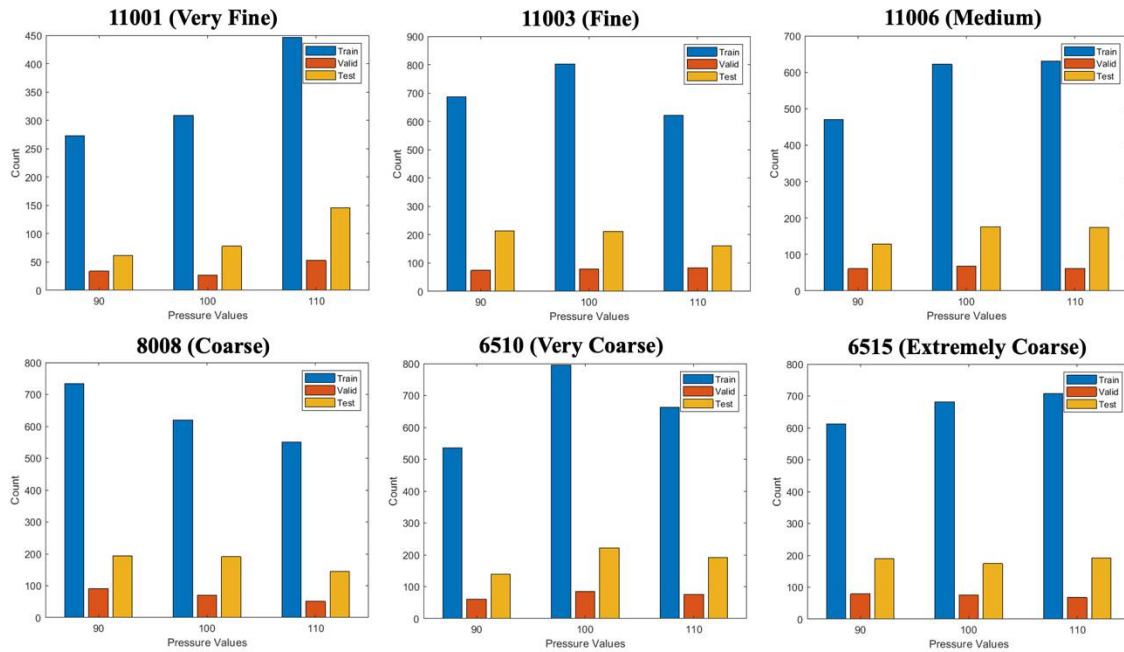


Figure 10 Allocation of images for training, testing, and validation for all six nozzle cases

V. Results

A. Pressure Identification from Shadowgraph Images

The mean absolute error (MAE) and the root mean squared error (RMSE) results for the pressure identification using the regression technique discussed in the previous section is shown in Figure 10 and Figure 11 for different nozzles from ROI 2 and ROI 3 as indicated in Figure 6, respectively. The error indicates the percentage within which nozzle pressure from the rated nozzle pressure was identified. For example, the rated pressure for a 11006 medium nozzle is 29 psi, as shown in Table 1, and it was sprayed at 29 psi, 26 psi, and 31 psi which corresponds to 100%, 90% and 110% of the rated pressure respectively. The results from Figure 10 indicates that the algorithm was able to identify the nozzle pressure with 3.12% error, which means that it could identify the pressure with an uncertainty of 29 ± 0.87 psi., 26 ± 0.78 psi, and 31 ± 0.93 psi, respectively. The error in predicting the pressure ranges from 3 to close to 5% of the applied pressure of the nozzle. The errors from different ROIs of the spray do not deviate significantly indicating the robustness of the algorithm in predicting the nozzle's applied pressure.

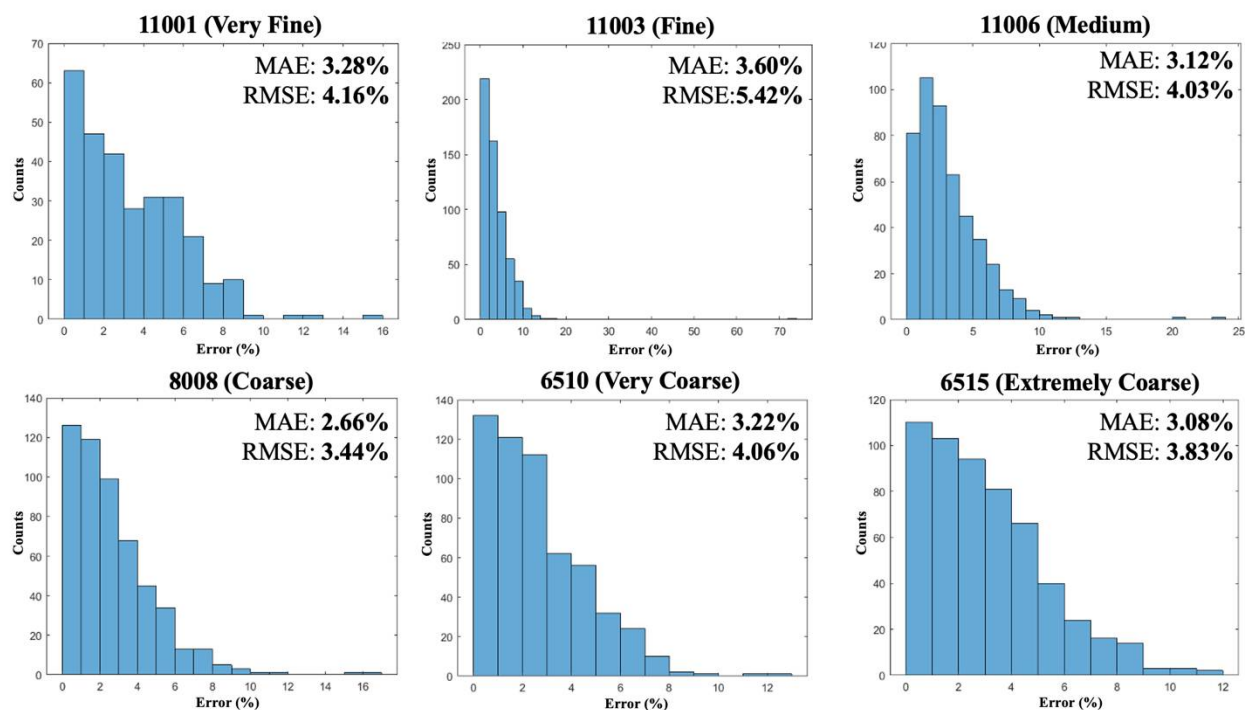


Figure 7 Mean absolute error (MAE) and Root mean squared error for pressure identification from different nozzles using the CNN approach for ROI II shown in Figure 6

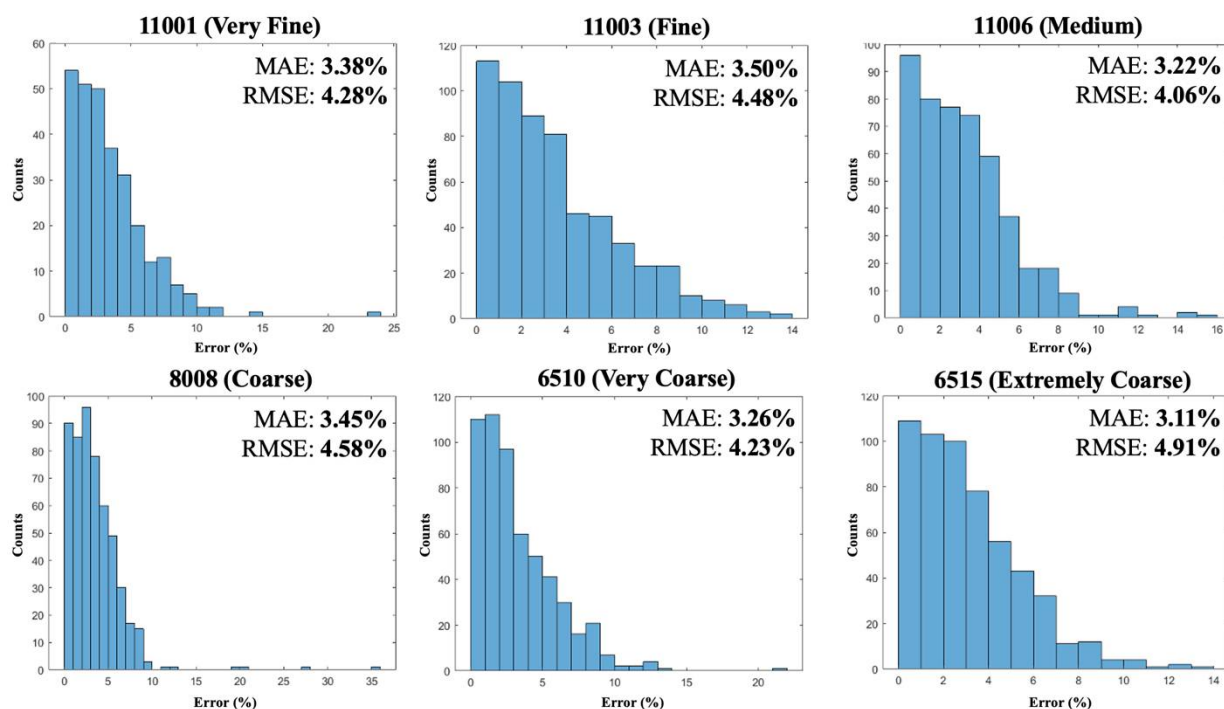


Figure 8 Mean absolute error (MAE) and Root mean squared error for pressure identification from different nozzles using the CNN approach for ROI III shown in Figure 6

B. Nozzle Identification using iPhone Images

The confusion matrix for the nozzle identification using images captured from iPhone 10 camera is shown in Figure 12. The results indicate that each nozzle was classified into their respective categories with 100% accuracy. Since there could be several factors that contributes to the higher accuracy in classification of the nozzle types, the class activation mapping (CAM) results were investigated to identify the fundamental basis behind the classification. CAM is a visualization tool where the probabilities predicted by the network are mapped back to the final convolutional layer to highlight the discriminative regions that are specific to that class. It is the result of the activation map of the ReLU (Rectified Linear Unit) after the last convolutional layer for a particular class which is determined by how much each activation contributes to the final probability score of that particular class. For an image, CAM represents the global average pooling layer applied following the last convolutional layer based on the spatial location to generate the weights. Therefore, it allows distinguishing the areas within an image that differentiates the class, which leads to its probability predictions. [15] Investigating the CAM images are important to ensure the classification is based on spray physics and atomization and not due to light intensity, nozzle angles, region of interest, etc. The CAM results from the CNN algorithm is shown in Figure 13 for all six nozzle types. The dark red regions in the CAM results indicate the regions of high probability that are discriminative to that nozzle type.

Confusion Matrix: Proposed CNN							
Output Class	Nozzle 11001	53 11.3%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Nozzle 11003	0 0.0%	51 10.9%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Nozzle 11006	0 0.0%	0 0.0%	160 34.3%	0 0.0%	0 0.0%	100% 0.0%
	Nozzle 6510	0 0.0%	0 0.0%	0 0.0%	65 13.9%	0 0.0%	100% 0.0%
	Nozzle 6515	0 0.0%	0 0.0%	0 0.0%	0 0.0%	72 15.4%	100% 0.0%
	Nozzle 8008	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	66 14.1% 100% 0.0%
		100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%
		Target Class					
		Nozzle 11001	Nozzle 11003	Nozzle 11006	Nozzle 6510	Nozzle 6515	Nozzle 8008

Figure 9 Confusion matrix for pressure identification using the CNN approach shown in Figure 8

It is interesting to note that the CAM is not focused on light intensity or the fan angles of the spray since 11001, 11003, and 11006 nozzles have the same fan angle and yet the CAM shows different focus regions that are particular to each nozzle type. Similar trends can be seen for the 6510 and 6515 nozzles where the fan angle is the same. Strong parallels can be drawn from comparing the CAM results from high speed shadowgraphy shown in Figure 3 to the CAM results from the iPhone camera.

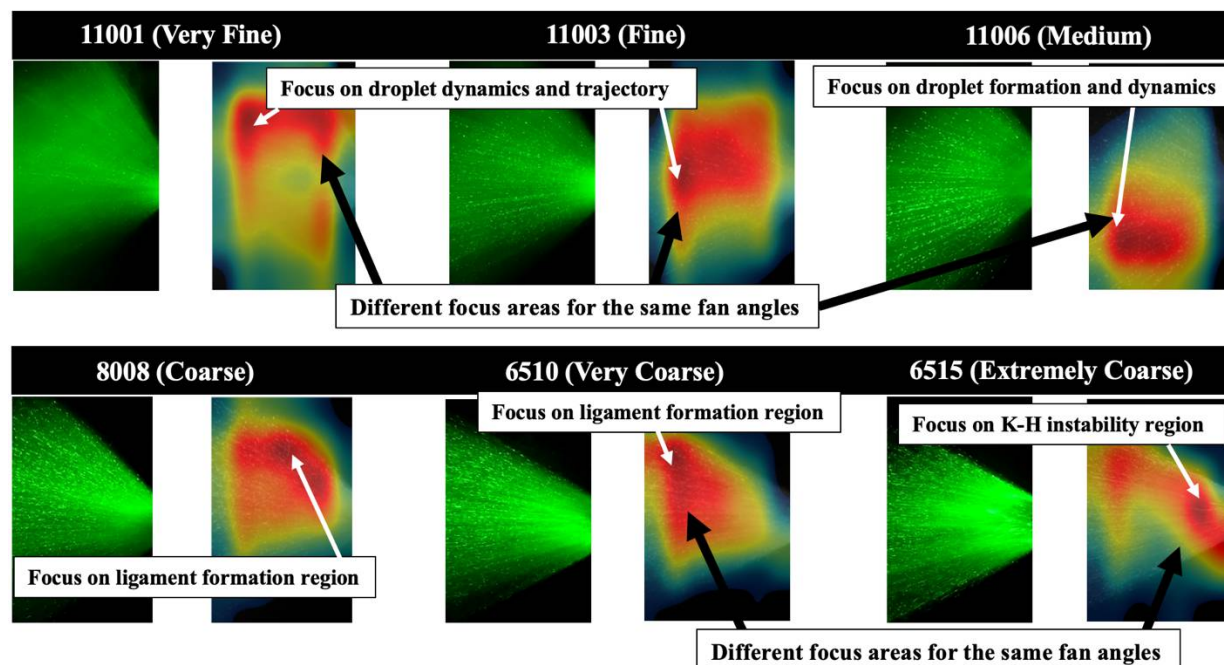


Figure 10 Spray image captured from iPhone 10 camera (left) and the corresponding mean CAM result for nozzle identification from the CNN (right)

For the extremely coarse nozzle, the CAM is focused on the region of the spray where Kelvin-Helmholtz instability takes place. On comparing the 6510 and the 8008 nozzle types, the CAM shows the focus region to the area where ligament formation occurs. At the same location for the CAM is higher for the 8008 nozzle, the droplets are well atomized for the 11006 nozzle since the K-H instability and ligament formation is pushed a closer to the nozzle when compared to 8008. The different atomization zones at the same distance from the nozzle for different types of nozzles is shown in Figure 14. For the 11006 and 11003 nozzles, the CAM is focused on the area where the spray is well atomized. In the case of 11001 nozzle, majority of the spray is dominated by smaller droplet diameter and hence the CAM is focused on majority of the spray region.

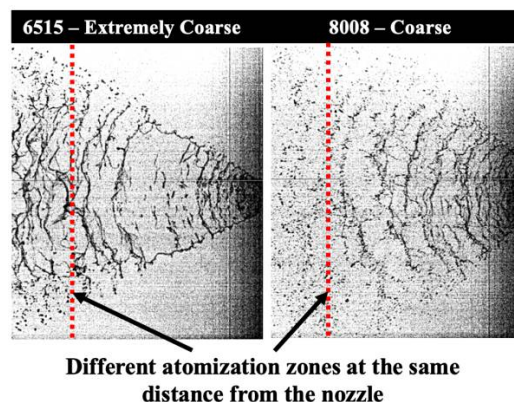


Figure 11 Different atomization zones at the same distance from 6515 and 8008 nozzles

VI. Conclusions

We extended our prior work on classifying spray pattern from different nozzle types and chemical compositions using convolutional neural network architecture to also identify the applied pressure at which the spray emanates from a given nozzle type and chemical composition. In this research, we presented a CNN architecture exclusive for this application. Custom CNN architecture performed well for both pressure estimation and nozzle identification illustrating its efficacy. The pressure identification was done using a simple CNN based regression approach and gave overall mean absolute error of 3-5% of the applied nozzle pressure. Since capturing spray images from a high-speed camera is not realistic to do by farmers in the field, an iPhone 10 camera was used to capture images of the spray from different nozzle types and we achieved an overall accuracy of 100%. Investigation of the mean CAM results of the classification algorithm indicated that the classification was based on atomization physics and not due to experimental setup, fan angle, or lighting conditions.

VII. References

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